

Revisiting QuantFormer for Hyperspectral Image Classification: Cross-Dataset Evaluation and Architectural Ablation

Abstract

QuantFormer [1] embeds a small variational quantum circuit as a spectral token encoder inside a shallow Vision Transformer for hyperspectral image classification, reporting over 99% accuracy on three airborne benchmarks using about 35,000 parameters. This project reproduces that architecture on Indian Pines and Pavia University, then extends it to Salinas, Kennedy Space Center, Botswana, and the full 15-class Houston 2013 scene, none of which the original paper evaluated. Pavia University matched the paper's reported accuracy within 1 percentage point; Indian Pines fell about 10 points short, a gap traced to per-class sample scarcity rather than an implementation error. A five-configuration ablation study across five datasets, with Benjamini-Hochberg-corrected, Fisher-combined McNemar significance testing, found that token-encoder bottleneck width, not the quantum-versus-classical choice, was the dominant factor in accuracy on most datasets. The quantum encoder versus classical mirror comparison was negligible to small on four datasets and large in magnitude on the remaining two, favoring the classical mirror on Indian Pines by 3.44 percentage points and favoring the quantum encoder on the full 15-class Houston 2013 scene, the hardest dataset tested, by 4.83 points, the largest margin in either direction observed in this project. A secondary experiment found that data augmentation helped more on datasets further from ceiling accuracy, and slightly hurt on the dataset closest to it. Embedding-separability metrics gave a mixed answer to the paper's claim that its quantum encoder produces a more discriminative feature space: the result depended on which metric and which dataset was examined.

1. Introduction

Hyperspectral image classification assigns a land-cover category to each pixel of a scene captured across dozens to hundreds of narrow, correlated spectral bands, typically with limited labeled training data relative to the number of pixels in a scene. QuantFormer [1] addresses this with a shallow transformer whose token encoder embeds a small variational quantum circuit: each spectral token, after PCA and a linear projection, is encoded as qubit rotation angles, processed through a shallow entangling circuit, and measured before being projected into the transformer's embedding space. The paper reports over 99% accuracy on three airborne benchmarks (Indian Pines, Pavia University, and a 7-class Houston 2013 subset) and 89.8% on EuroSAT_MS, using about 35,000 parameters. Its own ablation study finds that a classical projection of the same shape matches or slightly exceeds the quantum encoder's accuracy on all three datasets, while training two to three times faster; the paper does not claim a quantum accuracy advantage, only parameter efficiency and a controlled architectural comparison.

This project reproduces QuantFormer's architecture and training protocol on Indian Pines and Pavia University as a correctness checkpoint, then extends the identical pipeline to Salinas, Kennedy Space Center, Botswana, and the full 15-class Houston 2013 scene, none of which the original paper evaluated. The question is whether the paper's central finding, that the quantum encoder is not necessary for its reported accuracy, holds on datasets that are smaller, more class-imbalanced, or harder to classify than its

original three benchmarks. This documentation reports the reproduction, including a diagnosed but not fully resolved shortfall on Indian Pines; a five-configuration ablation study across five datasets with statistical testing; a data augmentation experiment; and a direct test, using embedding-separability metrics the original paper does not report, of QuantFormer's claim that its quantum encoder produces a more class-discriminative feature space.

2. Related Work / Background

Hyperspectral image classification has been approached with a range of deep learning architectures preceding QuantFormer, including convolutional networks that jointly model spectral and spatial structure, such as HybridSN [2] and the spectral-spatial residual network [5], and transformer-based approaches such as SpectralFormer [3]. Overall accuracy, average accuracy, Cohen's kappa, and confusion matrices are near-universal reported metrics in this literature, typically alongside a spatially rendered classification map showing predicted class per pixel across the full scene. This project follows that convention for its core metrics (Section 4) and adds classification maps for every dataset (Section 5). Receiver operating characteristic curves, precision-recall curves, calibration curves, and t-SNE embedding visualizations are less common in that literature and were included in this project specifically to test the original paper's own claim about embedding discriminability (Section 6.4), a claim the paper asserts but does not itself visualize or quantify.

3. Methodology

The pipeline, architecture, and training protocol described below follow the original QuantFormer specification [1] as closely as the paper's text permits. Where the paper does not state a value, that value is identified explicitly as a project decision rather than an extracted detail.

3.1 Locked Pipeline

Preprocessing followed a fixed order for every dataset: the labeled pixels were split first, normalization and PCA statistics were fit on the training split only, the fitted statistics were applied to transform all three splits, and spatial patches were extracted last. This order was enforced to prevent information from the validation or test splits from leaking into the fitted normalization or dimensionality-reduction statistics.

Each pixel's spectrum was normalized per band using min-max scaling, with the minimum and maximum computed from the training split only. Principal component analysis was then fit on the normalized training spectra, retaining components until cumulative explained variance reached 99%. A floor of 10 components was applied when the natural number of components needed to reach 99% variance fell below that value; every dataset except Indian Pines required this floor, a point discussed in Section 3.3.

A 15 by 15 spatial patch was extracted around every labeled pixel, using zero-padding at scene borders for Indian Pines and Pavia University and reflection padding for the four remaining datasets. Each patch was reshaped into 225 spectral tokens, one per spatial position, with each token a vector of length k equal to the number of retained PCA components for that dataset.

The quantum token encoder mapped each token to 4 rotation angles through a linear layer followed by a hyperbolic tangent scaled by π , then applied an RY angle embedding on 4 qubits followed by two layers of PennyLane's StronglyEntanglingLayers template [4]. The Pauli-Z expectation value of each qubit was measured, giving a 4-dimensional quantum feature vector, which a second linear layer projected to the transformer's embedding dimension of 64.

The transformer backbone consisted of a single encoder layer with an embedding dimension of 64, 4 attention heads, and a feed-forward dimension of 128, using a pre-layer-norm residual structure and a ReLU feed-forward activation. Fixed sinusoidal positional encoding was added to the 225 token embeddings before the encoder layer. The encoder's output was mean-pooled across all 225 tokens and passed to a single linear classification layer.

Training used the Adam optimizer with a learning rate of 2×10^{-3} , weight decay of 1×10^{-4} applied to classical weights only, a training batch size of 32, an evaluation batch size of 64, and 50 epochs per run. Gradients were clipped to a global norm of 1.0.

3.2 Hardware and Environment

All training was run on a single NVIDIA RTX A4500 GPU with 20 GB of dedicated VRAM, using Python 3.10.19, PyTorch 2.5.0 built against CUDA 11.8, and PennyLane 0.42.3.

Three quantum simulator backends were benchmarked before committing to a training configuration: PennyLane's default.qubit device with backpropagation differentiation, lightning.qubit with adjoint differentiation, and lightning.gpu with adjoint differentiation. On the locked workload (a training batch of 32 patches, 225 tokens each, processed through the 4-qubit circuit), default.qubit with backpropagation completed at 95.02 milliseconds per batch on CPU and at 59.10 milliseconds per batch when the same computation was run on CUDA tensors, a speedup of approximately 1.6 times. lightning.qubit with adjoint differentiation did not complete a 30-iteration timed benchmark within 17 minutes on the same workload and was excluded; the five warmup iterations for that backend completed successfully, indicating the slowdown was specific to the timed workload size rather than a failure to execute. lightning.gpu was not available in the installed PennyLane environment and was not benchmarked. default.qubit with backpropagation differentiation, executed on CUDA tensors, was used for every training run reported in this document.

A mechanical check confirmed that the quantum encoder processed each training batch through a single batched circuit invocation rather than one invocation per token: a batch of 32 patches (7,200 token vectors) produced one intercepted device execution rather than approximately 7,200.

3.3 Datasets

Indian Pines, Pavia University, Salinas, Kennedy Space Center, and Botswana used a stratified 10% training, 10% validation, and 80% test split, applied per class with a fixed split seed. This ratio is not stated in the original paper and is a project decision, chosen because it is a common convention in the hyperspectral classification literature. Houston 2013 used the official 2013 IEEE GRSS Data Fusion Contest train and test partition, with a stratified 10% of the official training portion held out as validation, since the official release does not include a separate validation split.

Five of the six datasets required the 10-component floor described in Section 3.1: Pavia University, Salinas, KSC, Botswana, and Houston 2013 each reached 99% cumulative PCA variance within 3 to 4 natural components. This was checked for a data quality artifact in each case (outlier-driven normalization and near-constant bands were examined) and found to be a genuine property of these five scenes rather than a defect in the normalization or PCA computation. Indian Pines did not require the floor, reaching 99% variance at 31 natural components, closer to the range of 15 to 30 components the original paper reports across its own four benchmarks.

The original paper defines a 7-class Houston 2013 subset with a stated scene size of 954 by 210 pixels and 48 spectral bands. These dimensions do not match the official Houston 2013 scene used in this project (349 by 1,905 pixels, 144 bands); they instead correspond to the documented overlap region between Houston 2013 and a separate, later Houston 2018 dataset, reduced to Houston 2018's 48-band sensor. Because the Houston 2018 dataset was not available for this project, the 7-class subset could not be reconstructed and was not evaluated. Only the full 15-class Houston 2013 scene is reported.

Dataset	Scene (H x W)	Bands	Classes	n_train	n_val	n_test	k_used
Indian Pines	145 x 145	200	16	1,024	1,025	8,200	31
Pavia University	610 x 340	103	9	4,277	4,277	34,222	10
Salinas	512 x 217	204	16	5,412	5,413	43,304	10
KSC	512 x 614	176	13	521	521	4,169	10
Botswana	1,476 x 256	145	14	324	324	2,600	10
Houston 2013 (15-class)	349 x 1,905	144	15	2,535	282	12,182	10

Table 1. Dataset splits and PCA dimensionality used in this project.

3.4 Ablation Design

Five model configurations were evaluated on every dataset except the dropped 7-class Houston subset:

- Full: the quantum token encoder described in Section 3.1, unchanged.
- w/o QNN: the quantum circuit replaced by a classical mirror of the same bottleneck shape, a linear layer from k to 4 dimensions, a hyperbolic tangent activation, and a linear layer from 4 to 64 dimensions.
- w/o Bottleneck: a single linear layer from k directly to 64 dimensions, with no 4-dimensional intermediate representation, used to separate the effect of the narrow bottleneck from the effect of the quantum-versus-classical choice.
- w/o PosEnc: the Full quantum encoder with the sinusoidal positional encoding removed.
- 1-Qubit: the quantum encoder reduced to a single qubit, using a linear layer from k to 1 dimension, RY embedding and StronglyEntanglingLayers on 1 qubit (which reduces to single-qubit rotation layers with no entanglement, since entanglement requires at least two qubits), and a linear layer from 1 to 64 dimensions.

3.5 Protocol Corrections

During development of the Indian Pines reproduction, two deviations from the intended training protocol were identified and corrected before any result in this document was finalized.

PyTorch's `nn.TransformerEncoderLayer` applies a default dropout of 0.1 unless explicitly overridden. This default was active in early runs without having been a deliberate protocol choice. On Indian Pines, with a training set of 1,024 patches and several classes represented by fewer than 10 training samples, this

dropout measurably reduced accuracy: a single seed's overall accuracy rose from 88.04% to 90.20% when dropout was set to 0.0, with no other change to the model or training loop. Dropout was set to 0.0 for all subsequent training.

The original training loop evaluated the model at the final epoch rather than selecting the checkpoint with the highest validation accuracy across training. Adding best-validation-checkpoint selection on top of the dropout correction raised the same seed's overall accuracy from 90.20% to 90.60%. Best-checkpoint selection was applied for all subsequent training.

A published-literature comparison found that convolutional methods commonly report 98% to 99% overall accuracy on Indian Pines using the same 10% training fraction used in this project, indicating that the training split size alone does not account for the gap between this reproduction and the original paper's reported 99.08% overall accuracy. This gap, and the diagnostic steps taken to investigate it, are reported in Section 5.1.1.

4. Experimental Setup

The full experimental matrix combines three blocks: a fidelity block (Indian Pines, Pavia University, and, originally, a Houston 2013 7-class subset that was later dropped for the reason given in Section 3.3), an extension-dataset Full-configuration block, and an extension-dataset ablation block. Table 2 gives the run count for each block as actually executed, after the Houston 7-class subset was removed from the fidelity block.

Block	Datasets	Seeds	Configurations	Augmentation	Runs
Fidelity	Indian Pines, Pavia University	3	5	None	30
Extension, Full config	Salinas, KSC, Botswana, Houston 2013 (15-class)	3	1	Both	24
Extension, ablation configs	Salinas, KSC, Botswana, Houston 2013 (15-class)	3	4	Unaugmented	48

Table 2. Executed run matrix by block.

This gives 30 fidelity runs plus 24 plus 48 extension runs, 102 runs in total. The project's locked protocol anticipated 117 runs across three fidelity datasets; the dropped Houston 7-class subset accounts for the 15-run difference (3 seeds times 5 configurations).

Three model seeds were used throughout: 42, 43, and 44. A separate, fixed split seed (1000) was used for the stratified train, validation, and test split on every dataset that used a stratified split, so that all three model seeds trained and evaluated on identical data and seed-to-seed variation reflects only initialization and batch order, not differences in which pixels were held out. Where data augmentation was applied, the augmentation random state was set to the model seed plus 1000, giving each seed a distinct but reproducible augmentation sequence.

Data augmentation, where applied, consisted of horizontal and vertical flips, a random rotation by 90, 180, or 270 degrees, and additive Gaussian noise with a standard deviation of 0.02. The geometric transforms were applied to the 15 by 15 spatial grid before tokenization, so that each token's position index continued to correspond to its raster position in the transformed grid and the fixed sinusoidal positional encoding remained consistent with the transformed layout. The noise component was applied to

the PCA-transformed token values rather than to the raw normalized spectral bands, because the raw normalized patches were not retained after the preprocessing stage; this is a deviation from the project's original protocol, which specified noise injection in raw band space, and is noted as a limitation in Section 9. Augmentation was applied only to the Full configuration on the four extension datasets and was not applied to any ablation configuration or to either fidelity dataset.

Every run recorded overall accuracy, average accuracy, Cohen's kappa, the Matthews correlation coefficient, a confusion matrix, per-class accuracy, and training time. Full-configuration runs on Indian Pines and Pavia University additionally recorded macro and weighted precision, recall, and F1, balanced accuracy, per-sample inference time, receiver operating characteristic and precision-recall curves, a confidence calibration curve, and, for those same two datasets, quantum and classical token embeddings used for a t-SNE projection, a silhouette score, a Fisher discriminant ratio, and a linear-probe classification accuracy. This tiering, full evaluation suite on the two fidelity datasets and core metrics elsewhere, followed the project's original protocol, which reserved the more expensive analyses for the datasets serving as correctness checkpoints.

Statistical comparisons between the Full configuration and each ablation configuration used McNemar's test on paired predictions from the same test set, computed separately for each of the three seeds. The three seeds' p-values for each comparison were corrected for multiple comparisons within each seed using the Benjamini-Hochberg procedure across the four ablation comparisons, then combined across seeds using Fisher's method to produce one combined p-value per dataset per comparison.

Because model checkpoints were not retained between the extraction and evaluation stages of this project, the Full configuration at seed 42 was retrained independently at several points: once for the headline reproduction or extension result reported in Section 5, once as the ablation-study baseline reported in Section 6, and, for Indian Pines and Pavia University, once more to extract the class probabilities and embeddings used for the receiver operating characteristic, precision-recall, calibration, and t-SNE figures. These retrains used identical code, architecture, and seed, but produced slightly different final accuracy (for example 88.38% in the original Indian Pines run against 88.54% in the retrain used for those figures), a difference attributable to GPU floating-point non-determinism rather than to any methodological change. All such differences observed in this project were under one percentage point and fall within the seed-to-seed variation already reported for each dataset. Confusion matrices in Section 5 use the original headline run for each dataset; classification maps, and the receiver operating characteristic, precision-recall, calibration, and t-SNE figures for Indian Pines and Pavia University, use the separate retrain performed for artifact extraction.

5. Results

Indian Pines and Pavia University reproduce the original paper's exact protocol and are compared directly against its reported accuracy. Salinas, Kennedy Space Center, Botswana, and the full 15-class Houston 2013 scene were not evaluated in the original paper and are reported as this project's extension.

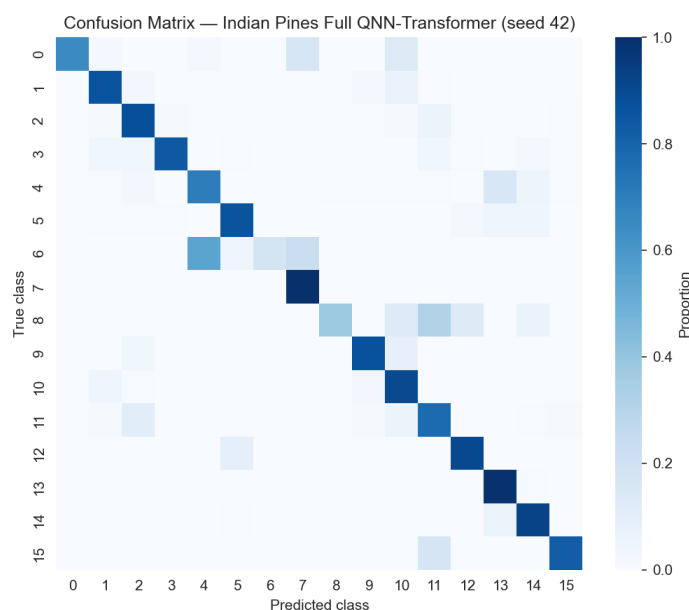
5.1 Fidelity Checkpoints

5.1.1 Indian Pines

Table 3 gives the overall accuracy, average accuracy, and Cohen's kappa for the two individual seeds whose full metrics were recorded in this project's session log; the third seed's individual average accuracy and kappa were not captured in that log, but its overall accuracy contributed to the three-seed mean overall accuracy of 88.97%, which corresponds to a gap of 10.11 percentage points from the paper's reported 99.08%.

Metric	Seed 42	Seed 43	3-seed mean OA	Paper
Overall accuracy	88.38%	90.28%	88.97%	99.08%
Average accuracy	78.51%	81.40%	not recorded	98.86%
Cohen's kappa	0.8673	0.8892	not recorded	0.990

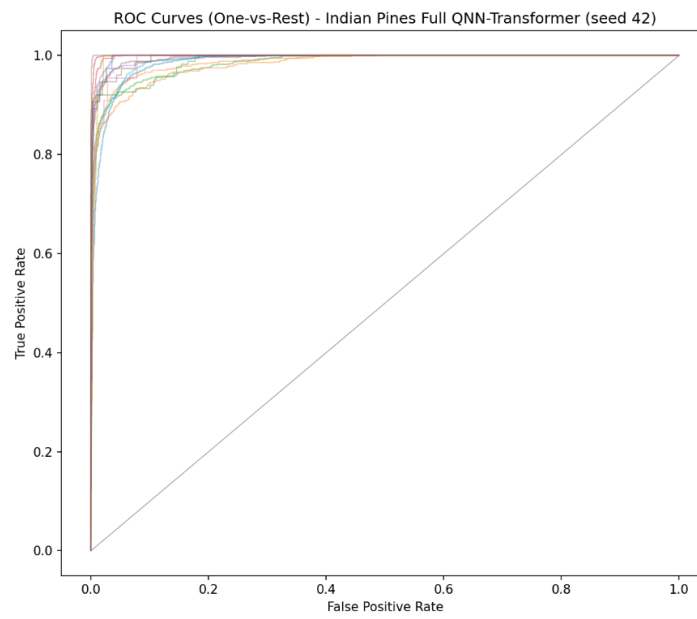
Table 3. Indian Pines results against the paper's reported values.



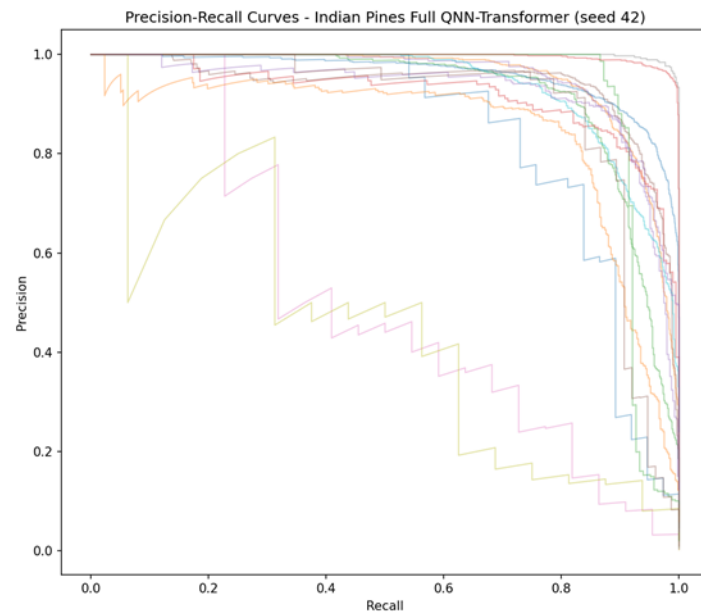
Confusion matrix, Indian Pines, Full configuration, seed 42



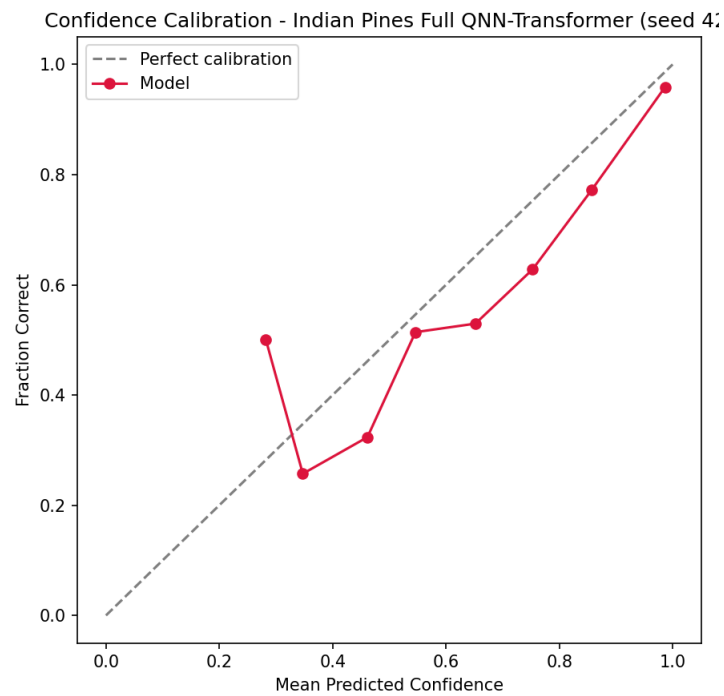
Classification map, Indian Pines, Full Configuration, seed 42



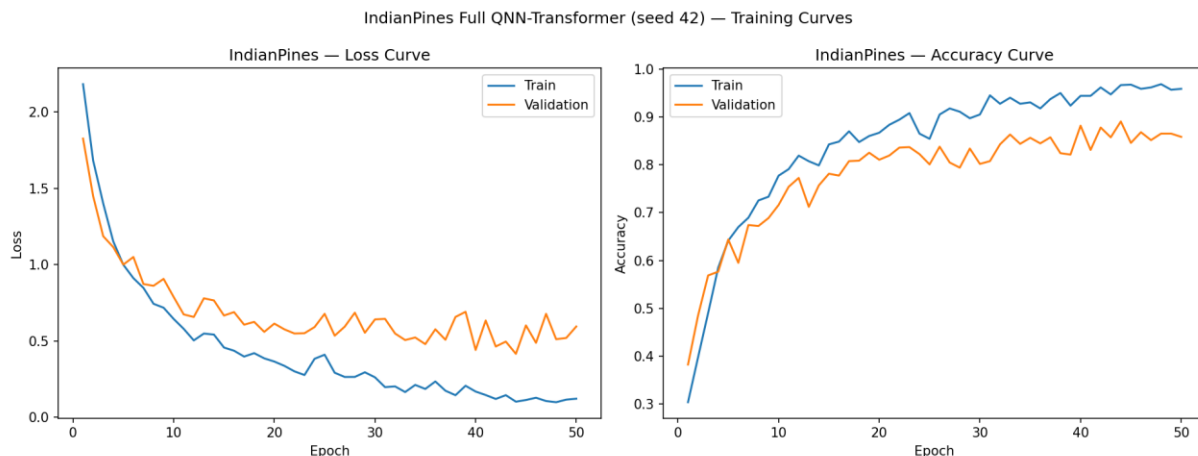
Receiver operating characteristic curves, one-vs-rest, Indian Pines, Full configuration, seed 42.



Precision-recall curves, one-vs-rest, Indian Pines, Full configuration, seed 42.



Confidence calibration curve, Indian Pines, Full configuration, seed 42.



Training and validation loss and accuracy curves, Indian Pines, Full configuration, seed 42.

A per-class check on the seed-42 run found that overall accuracy is not evenly distributed across classes. The two classes with the fewest training samples had the lowest test accuracy: class 8, with 2 training samples, reached 37.5% test accuracy, and class 6, with 3 training samples, reached 59.1%. Sample count alone did not explain the full pattern, however; a class with only 5 training samples (class 0) reached 83.8% test accuracy, higher than several classes with 40 to 60 training samples, so per-class sample scarcity is a contributing factor rather than the sole explanation for the shortfall against the paper.

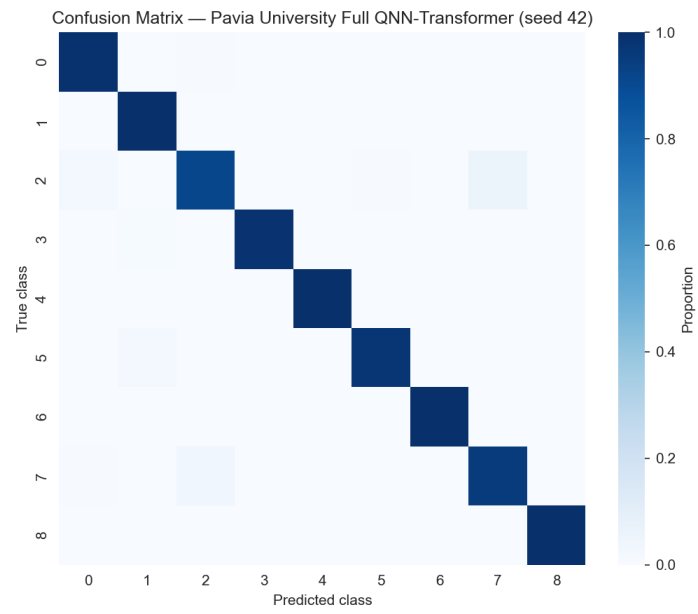
A literature comparison found that convolutional methods commonly reach 98% to 99% overall accuracy on Indian Pines using the same 10% training fraction used in this project (Section 3.5), which argues against training-set size alone as the explanation. Removing an unintended default dropout and adding best-validation-checkpoint selection (Section 3.5) recovered part of the gap (a single seed's accuracy rose from 88.04% to 90.60%) but did not close it. The original paper provides no public code release and does not report per-seed variance, so the remaining gap cannot be attributed with confidence to either an unresolved difference in this implementation or to a less conservative aspect of the paper's own evaluation, such as single-seed reporting or a train-test split with greater spatial overlap between patches.

5.1.2 Pavia University

Table 4 gives the same breakdown for Pavia University. The three-seed mean overall accuracy of 98.99% is within 1 percentage point of the paper's reported 99.94%, meeting this project's fidelity threshold. The paper's text used for this project reports only overall accuracy for Pavia University; average accuracy and Cohen's kappa are not available for comparison.

Metric	Seed 42	Seed 43	3-seed mean OA	Paper
Overall accuracy	98.45%	99.49%	98.99%	99.94%
Average accuracy	97.92%	99.05%	not recorded	not available
Cohen's kappa	0.9795	0.9932	not recorded	not available

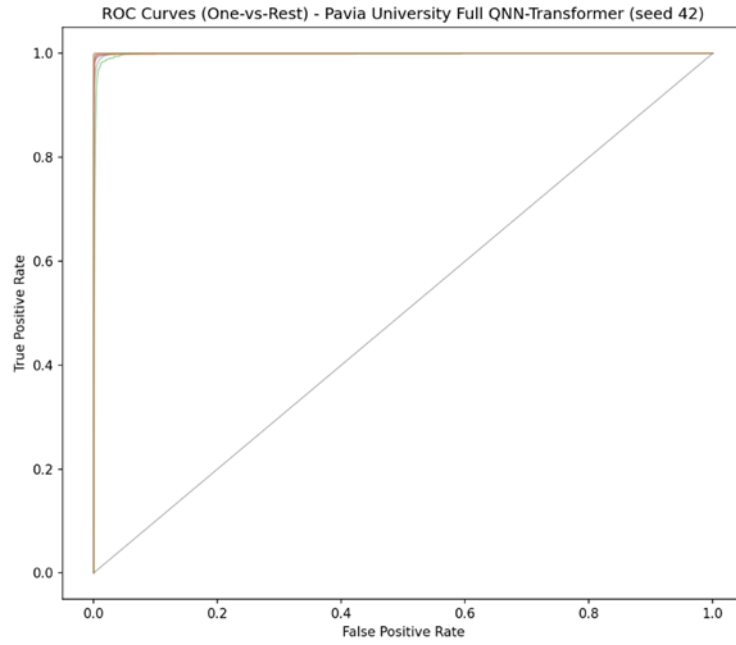
Table 4. Pavia University results against the paper's reported values.



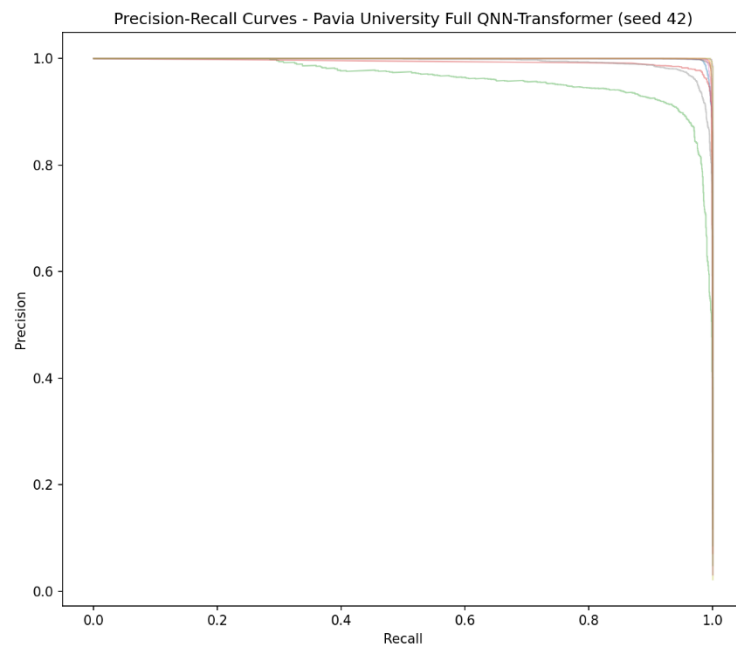
Confusion matrix, Pavia University, Full configuration, seed 42.



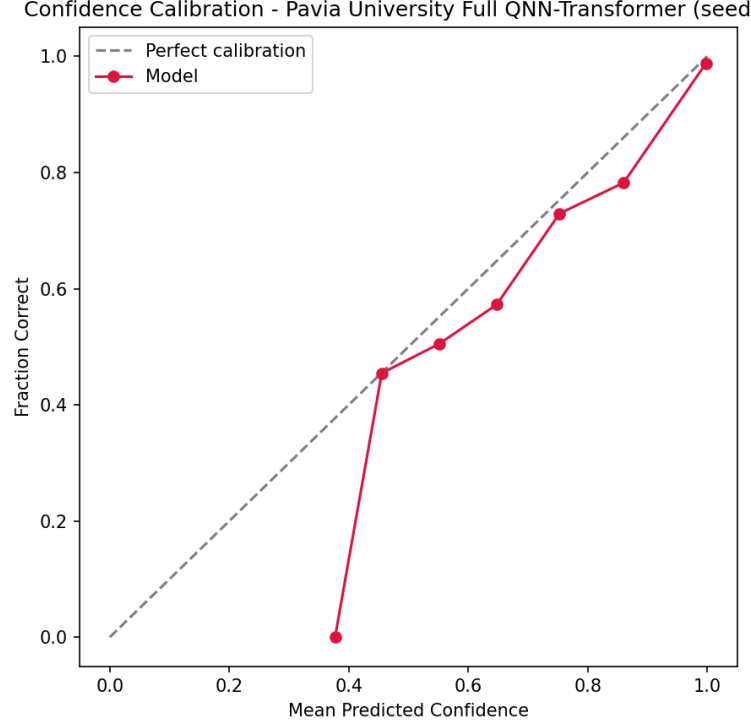
Classification map, Pavia University, Full configuration, seed 42.



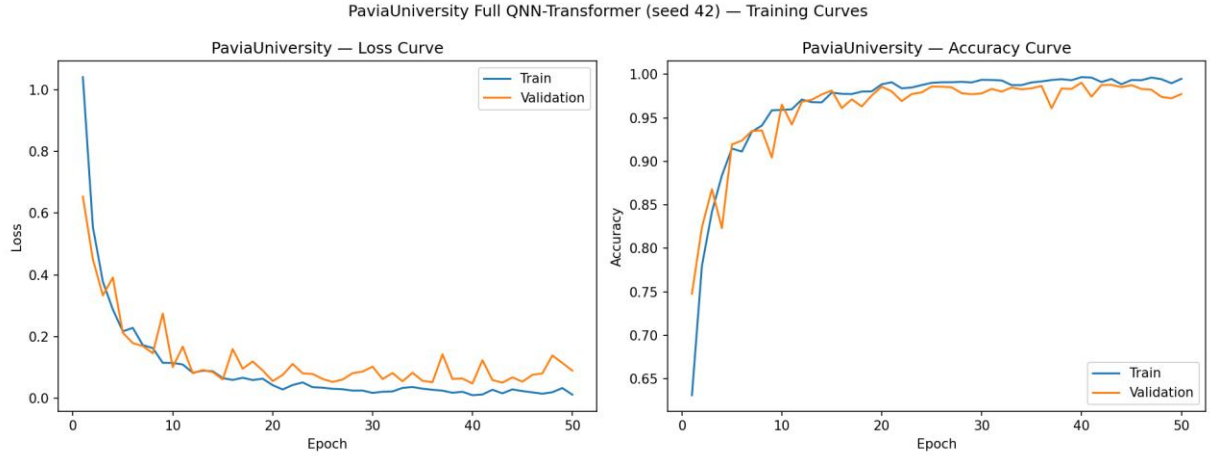
Receiver operating characteristic curves, one-vs-rest, Pavia University, Full configuration, seed 42.



Precision-recall curves, one-vs-rest, Pavia University, Full configuration, seed 42.



Confidence calibration curve, Pavia University, Full configuration, seed 42.



Training and validation loss and accuracy curves, Pavia University, Full configuration, seed 42.

Pavia University's close match to the paper's reported accuracy, alongside Indian Pines' unresolved gap, is consistent with the per-class evidence in Section 5.1.1: Pavia University's training split (4,277 patches across 9 classes) does not contain the severely undersized classes present in Indian Pines' split (several classes under 10 training samples out of 1,024 total), and this dataset's implementation passes the fidelity check.

5.2 Extension Datasets

The four extension datasets were each evaluated with three seeds, both without and with data augmentation, using the Full configuration. Table 5 gives the three-seed mean and standard deviation of overall accuracy for each.

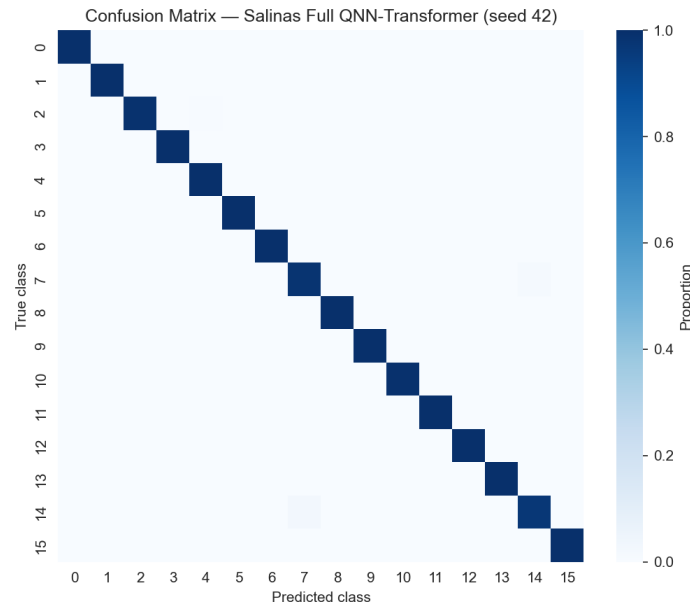
Dataset	OA, unaugmented	OA, augmented	Augmentation effect
Salinas	99.26% (SD 0.22)	98.90% (SD 0.05)	-0.36 pp
KSC	96.47% (SD 0.99)	97.56% (SD 0.53)	+1.09 pp
Botswana	94.49% (SD 1.00)	96.08% (SD 1.60)	+1.59 pp
Houston 2013 (15-class)	74.19%	77.26%	+3.08 pp

Table 5. Extension dataset results, unaugmented versus augmented, three-seed mean.

Standard deviation across seeds is given where it was recorded; individual per-seed overall accuracy for Houston 2013 was 74.20%, 74.80%, and 73.57% unaugmented, and 77.91%, 77.80%, and 76.08% augmented, from which the means above are computed.

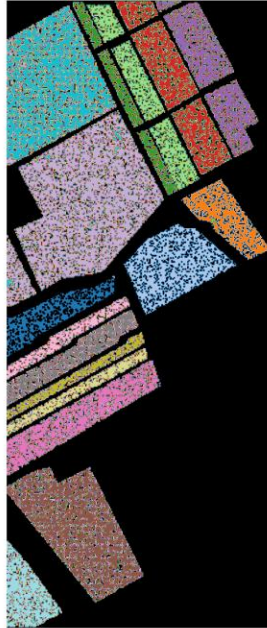
5.2.1 Salinas

Salinas has the largest training split of the four extension datasets (5,412 patches) and the highest unaugmented accuracy (99.26%). Augmentation produced a small decrease here (-0.36 percentage points) and a lower standard deviation across seeds, the only extension dataset where augmentation did not help.



Confusion matrix, Salinas, Full configuration, seed 42.

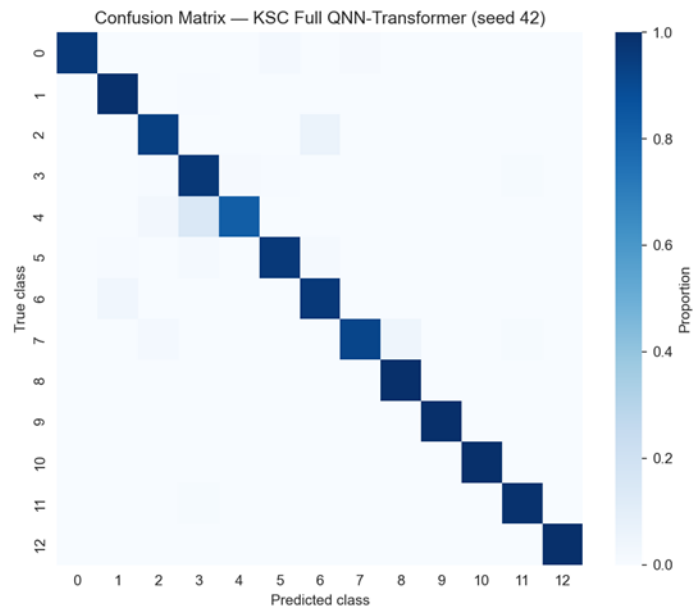
Classification Map — Salinas Full QNN-Transformer (seed 42)



Classification map, Salinas, Full configuration, seed 42.

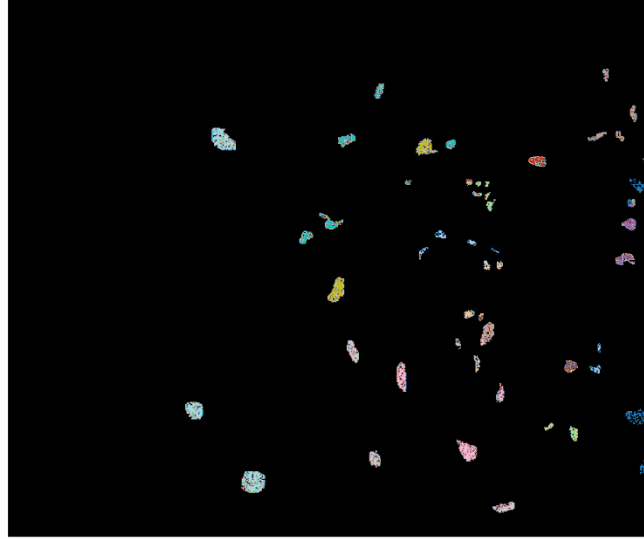
5.2.2 KSC

Kennedy Space Center has 521 training patches across 13 classes. Augmentation improved accuracy by 1.09 percentage points.



Confusion matrix, KSC, Full configuration, seed 42.

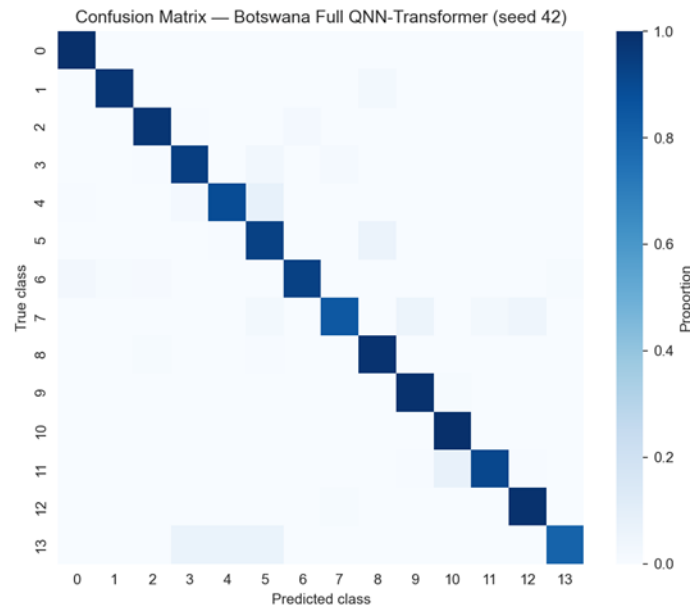
Classification Map — KSC Full QNN-Transformer (seed 42)



Classification map, KSC, Full configuration, seed 42.

5.2.3 Botswana

Botswana has the smallest training split of the four extension datasets (324 patches across 14 classes) and the lowest unaugmented accuracy among the three original extension datasets (94.49%). Augmentation improved accuracy by 1.59 percentage points, the largest improvement among Salinas, KSC, and Botswana. A spatially disjoint train-test split probe was planned for this dataset as an additional robustness check but was not run in this project; this is noted as a limitation in Section 9.



Confusion matrix, Botswana, Full Configuration, seed 42

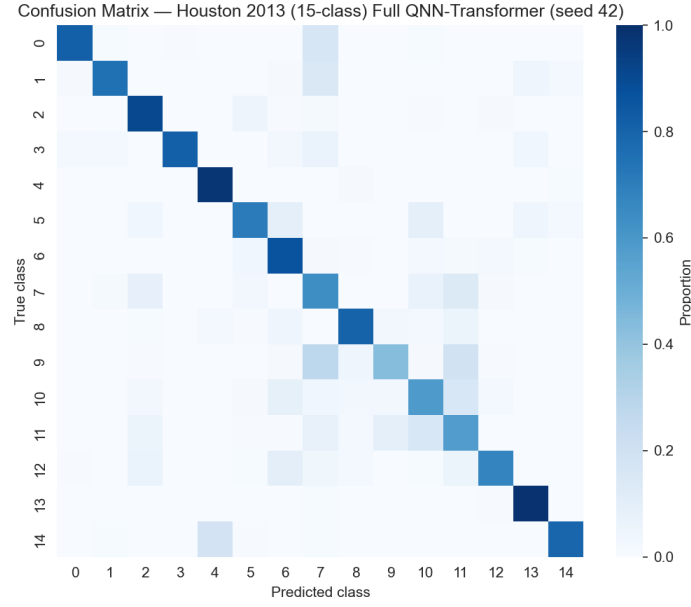
Classification Map — Botswana Full QNN-Transformer (seed 42)



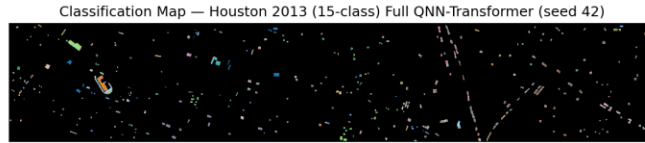
Classification map, Botswana, Full configuration, seed 42.

5.2.4 Houston 2013 (15-class)

The full 15-class Houston 2013 scene reached the lowest overall accuracy of any dataset in this project, 74.19% unaugmented and 77.26% augmented, the largest augmentation improvement observed (+3.08 percentage points). Average accuracy and Cohen's kappa, computed from the three individual seed values recorded for the unaugmented run, were 75.50% and 0.7203 respectively. The 7-class Houston subset the original paper evaluates could not be reconstructed from the data available to this project, for the reason given in Section 3.3, and is not reported. Section 6.3 examines this dataset's ablation results in more detail, since its pattern differs from the other five datasets.



Confusion matrix, Houston 2013 (15-class), Full configuration, unaugmented, seed 42.



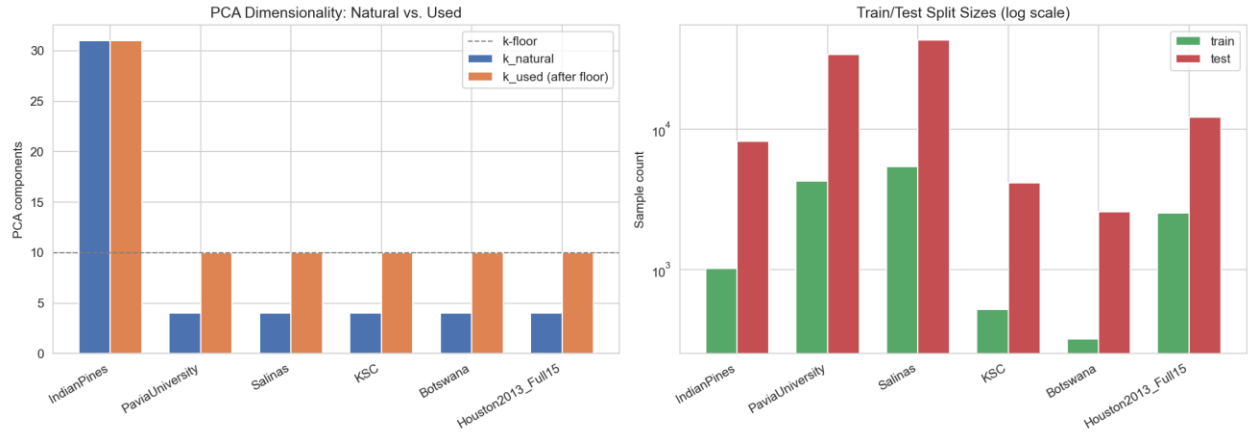
Classification map, Houston 2013 (15-class), Full configuration, seed 42.

5.3 Cross-Dataset Comparison

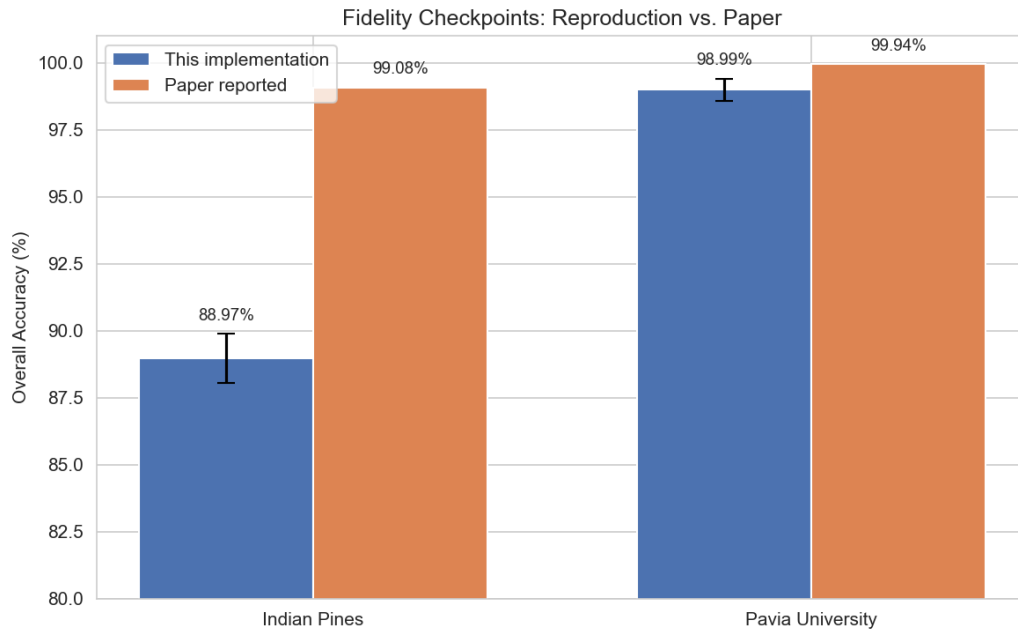
Table 6 compares overall accuracy, macro F1, and the Matthews correlation coefficient across all six datasets, using the Full-configuration results from the ablation study (Section 6) as a single consistent source. These are separate training runs from the headline results in Sections 5.1 and 5.2 and can differ from them by less than one percentage point for the reason given in Section 4.

Dataset	OA (Full, 3-seed mean)	MCC (Full, 3-seed mean)
Indian Pines	88.58%	0.8700
Pavia University	98.92%	0.9857
Salinas	99.28%	0.9920
KSC	96.47%	0.9608
Botswana	94.49%	0.9405
Houston 2013 (15-class)	74.19%	0.7223

Table 6. Overall accuracy and Matthews correlation coefficient, all six datasets, Full configuration.



PCA dimensionality and train/test split sizes, all six datasets.



Fidelity checkpoint comparison against paper-reported accuracy, Indian Pines and Pavia University.

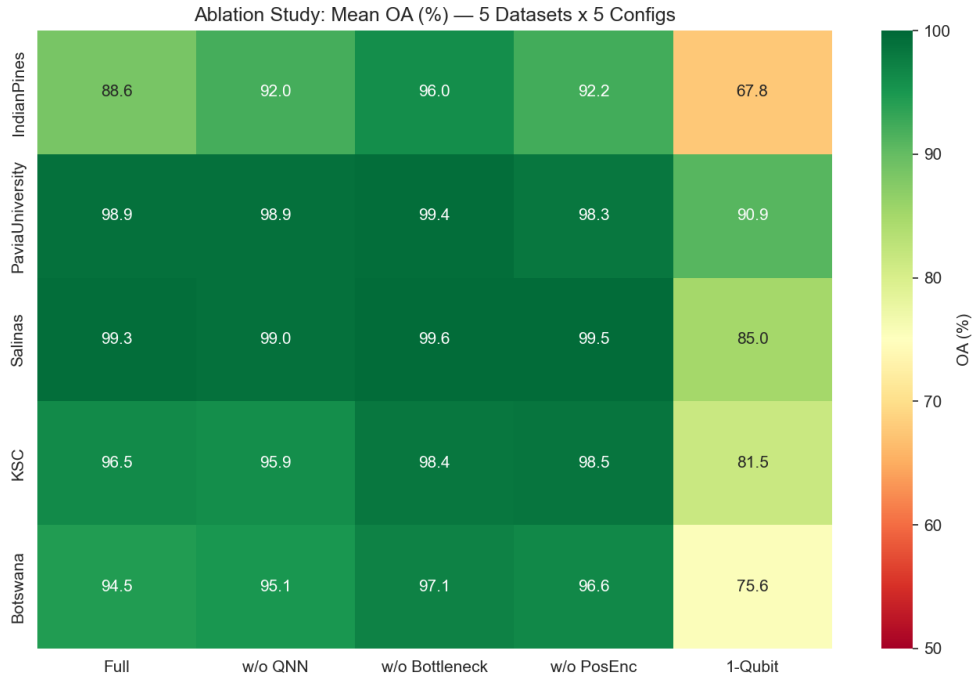
6. Ablation Study

6.1 Full Ablation Matrix

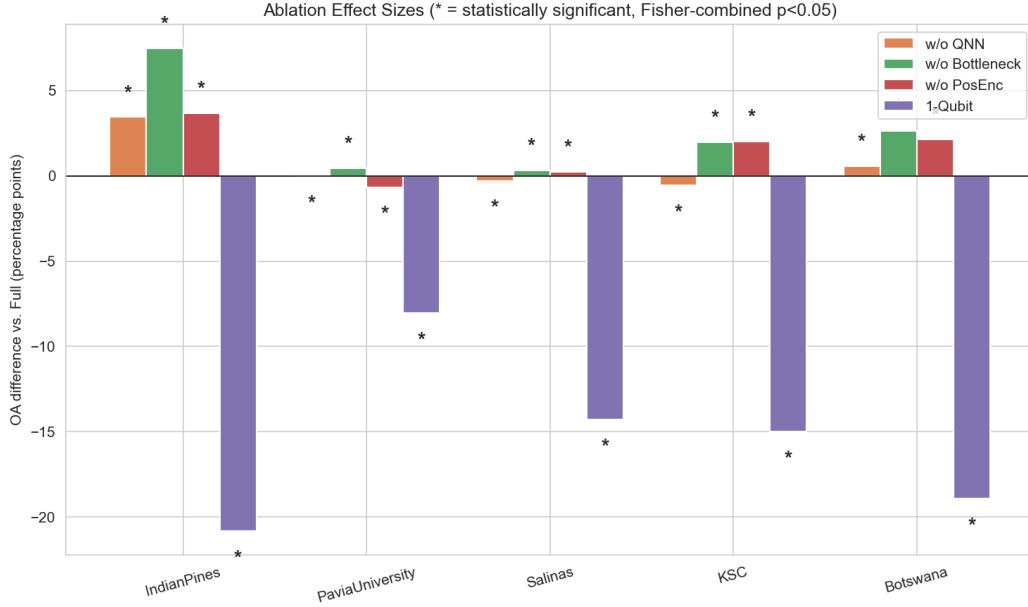
Table 7 gives the three-seed mean overall accuracy for all five configurations on the five datasets that received the full ablation suite (the Houston 7-class subset was dropped before this stage, as described in Section 3.3; the Houston 15-class scene's ablation results are reported separately in Section 6.3).

Dataset	Full	w/o QNN	w/o Bottleneck	w/o PosEnc	1-Qubit
Indian Pines	88.58%	92.02%	96.04%	92.25%	67.76%
Pavia University	98.92%	98.90%	99.37%	98.25%	90.87%
Salinas	99.28%	99.00%	99.59%	99.50%	85.00%
KSC	96.47%	95.91%	98.43%	98.50%	81.47%
Botswana	94.49%	95.06%	97.12%	96.62%	75.56%

Table 7. Three-seed mean overall accuracy, five datasets by five configurations.



Mean overall accuracy, five datasets by five configurations.



Ablation effect sizes relative to Full configuration, with significance markers.

w/o Bottleneck reached the highest or tied-highest accuracy on every one of the five datasets, with the largest gain on Indian Pines (+7.46 percentage points over Full). 1-Qubit reached the lowest accuracy on every dataset, with losses ranging from -8.04 points on Pavia University to -20.82 points on Indian Pines. w/o QNN, the direct quantum-versus-classical comparison, produced small effects on four of the five datasets (-0.56 to +0.58 percentage points) and a larger effect only on Indian Pines (+3.44 points, favoring the classical mirror).

6.2 Statistical Testing

Table 8 gives the effect size (the ablation configuration's accuracy minus Full's accuracy, in percentage points) and the combined, multiple-comparison-corrected McNemar significance for each dataset and comparison. Effect sizes under 0.5 percentage points are labeled negligible, 0.5 to 3 points moderate, and above 3 points large.

Dataset	Comparison	Effect (pp)	Magnitude	Combined p	Significant
Indian Pines	w/o QNN	+3.44	Large	1.5e-50	Yes
Indian Pines	w/o Bottleneck	+7.46	Large	1.2e-274	Yes
Indian Pines	w/o PosEnc	+3.67	Large	2.9e-68	Yes
Indian Pines	1-Qubit	-20.82	Large	<1e-300	Yes
Pavia University	w/o QNN	-0.02	Negligible	9.9e-27	Yes
Pavia University	w/o Bottleneck	+0.46	Negligible	6.6e-54	Yes
Pavia University	w/o PosEnc	-0.67	Moderate	7.0e-67	Yes
Pavia University	1-Qubit	-8.04	Large	<1e-300	Yes
Salinas	w/o QNN	-0.28	Negligible	8.6e-36	Yes
Salinas	w/o Bottleneck	+0.31	Negligible	1.2e-28	Yes
Salinas	w/o PosEnc	+0.22	Negligible	1.8e-18	Yes
Salinas	1-Qubit	-14.28	Large	<1e-300	Yes
KSC	w/o QNN	-0.56	Moderate	2.9e-4	Yes
KSC	w/o Bottleneck	+1.96	Moderate	9.9e-31	Yes
KSC	w/o PosEnc	+2.02	Moderate	4.8e-27	Yes
KSC	1-Qubit	-15.00	Large	<1e-300	Yes
Botswana	w/o QNN	+0.58	Moderate	0.048	Yes (borderline)

Botswana	w/o Bottleneck	+2.63	Moderate	3.9e-19	Yes
Botswana	w/o PosEnc	+2.13	Moderate	2.3e-13	Yes
Botswana	1-Qubit	-18.92	Large	7.4e-255	Yes

Table 8. McNemar effect size and significance, Full versus each ablation configuration.

Every comparison in Table 8 reached statistical significance after correction, including several with negligible effect sizes (for example Pavia University's w/o QNN comparison, a 0.02-point difference). This reflects the size of the test sets involved (up to 43,304 test pixels for Salinas) rather than practical importance: McNemar's test, like most significance tests, becomes highly sensitive to small differences at large sample sizes. The effect-size and magnitude columns in Table 8 should be read alongside significance rather than in place of it; the negligible-magnitude, statistically significant results indicate that Full and the compared configuration produce detectably different predictions without indicating that one is meaningfully more accurate.

6.3 Houston-15 Detailed Analysis

The full 15-class Houston 2013 scene was the only dataset in this project where Full outperformed every ablation configuration. Table 9 gives the effect sizes and significance for this dataset, computed against the unaugmented Full configuration.

Comparison	Effect (pp)	Magnitude	Combined p	Significant
w/o QNN	-4.83	Large	9.2e-96	Yes
w/o Bottleneck	-0.63	Moderate	8.8e-19	Yes
w/o PosEnc	-1.22	Moderate	1.8e-8	Yes
1-Qubit	-17.09	Large	<1e-300	Yes

Table 9. McNemar effect size and significance, Houston 2013 (15-class), Full versus each ablation configuration.

The quantum-versus-classical comparison (w/o QNN) shows the quantum encoder ahead by 4.83 percentage points, its largest advantage over the classical mirror on any dataset in this project; on every other dataset that comparison favored the classical mirror or was negligible.

A per-class check across all three seeds found this advantage was not spread evenly across Houston's 15 classes. Two classes showed a consistent pattern across seeds: class 14 favored the quantum encoder in all three seeds (a difference of +56.8, +81.8, and +2.3 percentage points), and class 5 favored the classical mirror in all three seeds (-21.1, -12.7, and -5.6 points). Two other classes did not show a consistent pattern: class 2 favored the quantum encoder in two of three seeds and the classical mirror in the third (+43.1, -30.8, +25.8 points), and class 11 showed a similar inconsistency (-19.0, +4.7, -19.6 points). The dataset's overall quantum advantage is therefore driven by a subset of classes rather than a uniform effect, and two of the four classes examined individually did not show a seed-stable direction.

Houston 2013 is also the hardest classification problem in this project by overall accuracy (74.19% against 94.49% to 99.28% on the other five datasets). Its results are consistent with a possible relationship between problem difficulty and the quantum encoder's measured contribution, though this project tested that relationship on only one hard dataset and cannot establish it as a general pattern.

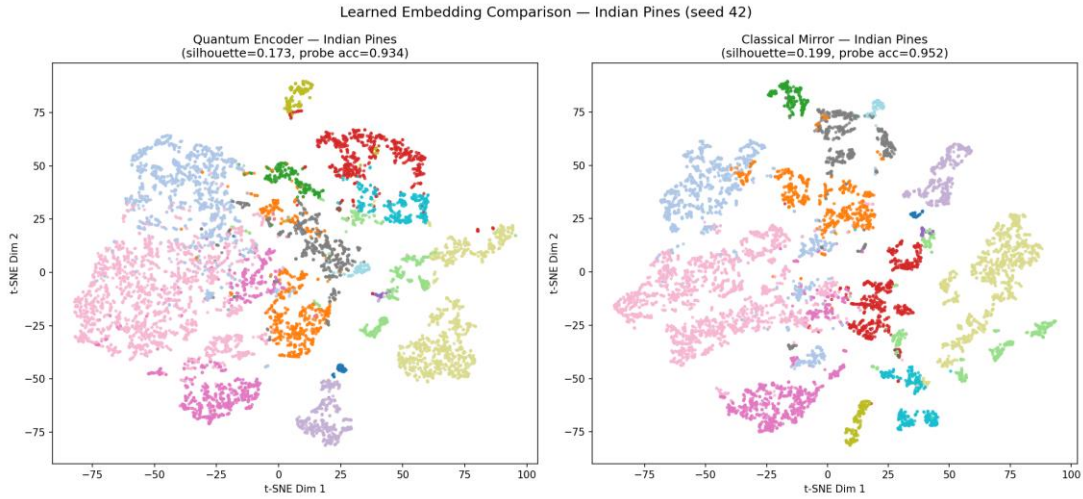
6.4 Embedding Separability (Quantum vs. Classical)

For Indian Pines and Pavia University, quantum and classical token embeddings were extracted from the pre-classifier representation of a seed-42 Full-configuration model and its w/o QNN counterpart, and

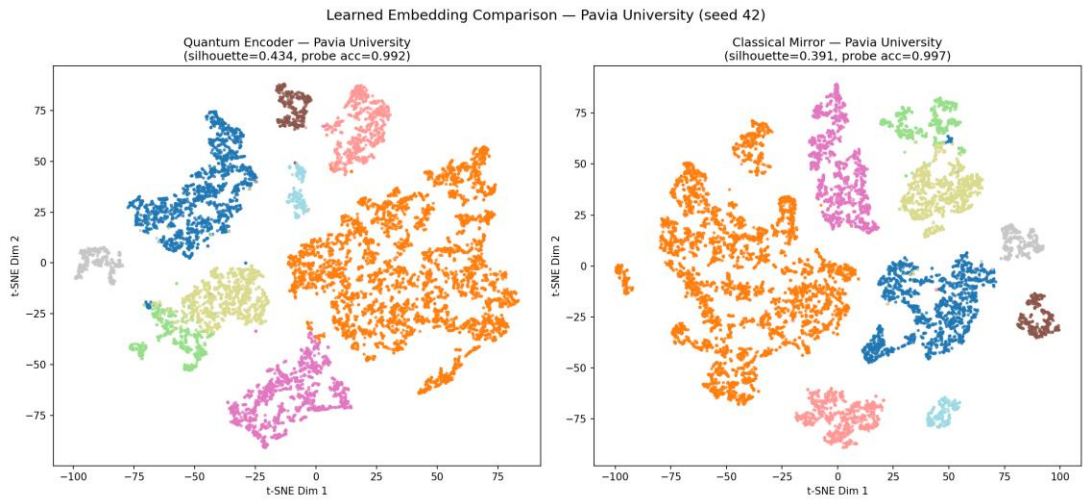
compared using three separability metrics: the silhouette score, a Fisher discriminant ratio (the ratio of between-class to within-class variance, averaged across embedding dimensions), and the accuracy of a logistic regression classifier trained on the frozen embeddings. Table 10 gives the results.

Dataset	Metric	Quantum	Classical mirror
Indian Pines	Silhouette score	0.1733	0.1987
Indian Pines	Fisher discriminant ratio	3.1751	3.1753
Indian Pines	Linear-probe accuracy	93.35%	95.24%
Pavia University	Silhouette score	0.4340	0.3911
Pavia University	Fisher discriminant ratio	4.9725	4.3605
Pavia University	Linear-probe accuracy	99.18%	99.69%

Table 10. Embedding separability metrics, quantum encoder versus classical mirror.



t-SNE projection of quantum and classical mirror embeddings, Indian Pines, seed 42.



t-SNE projection of quantum and classical mirror embeddings, Pavia University, seed 42.

The result does not uniformly favor either encoder. On Pavia University, the quantum encoder produced a higher silhouette score and a higher Fisher discriminant ratio than the classical mirror; on Indian Pines it did not. On both datasets, the classical mirror's embeddings supported higher linear-probe accuracy. This indicates that the original paper's claim of a more class-discriminative quantum feature space is not

supported uniformly: the quantum embedding space was more geometrically clustered by two of three metrics on one dataset, while producing a space that was less linearly separable on both datasets tested.

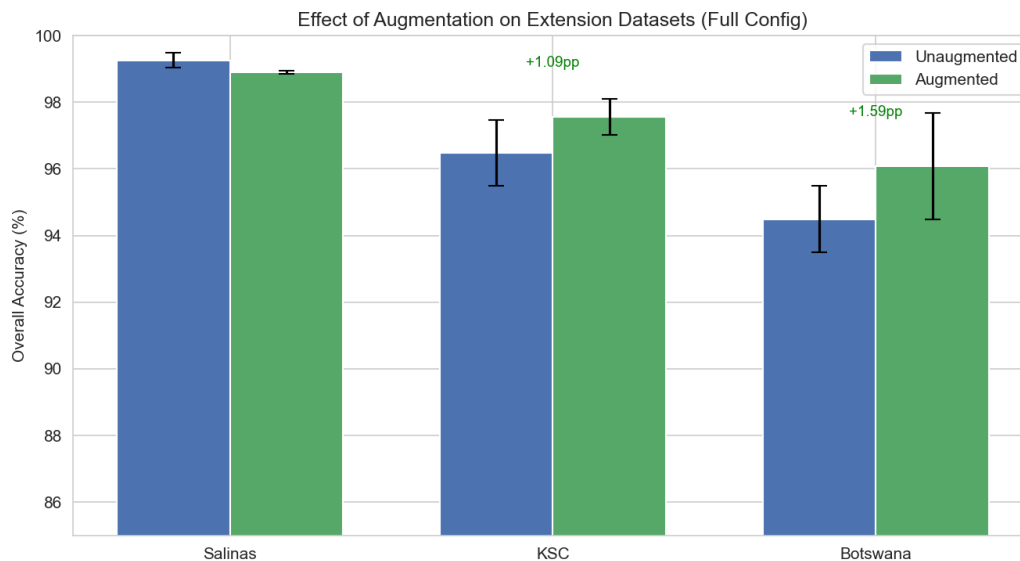
An activation-function sensitivity check for the w/o QNN classical mirror (comparing its tanh activation, chosen to mirror the quantum encoder's own angle-embedding nonlinearity, against ReLU or GELU) was planned to test whether this choice affects the classical mirror's reported capacity, but was not run in this project; this is noted as a limitation in Section 9.

6.5 Augmentation Effect Summary

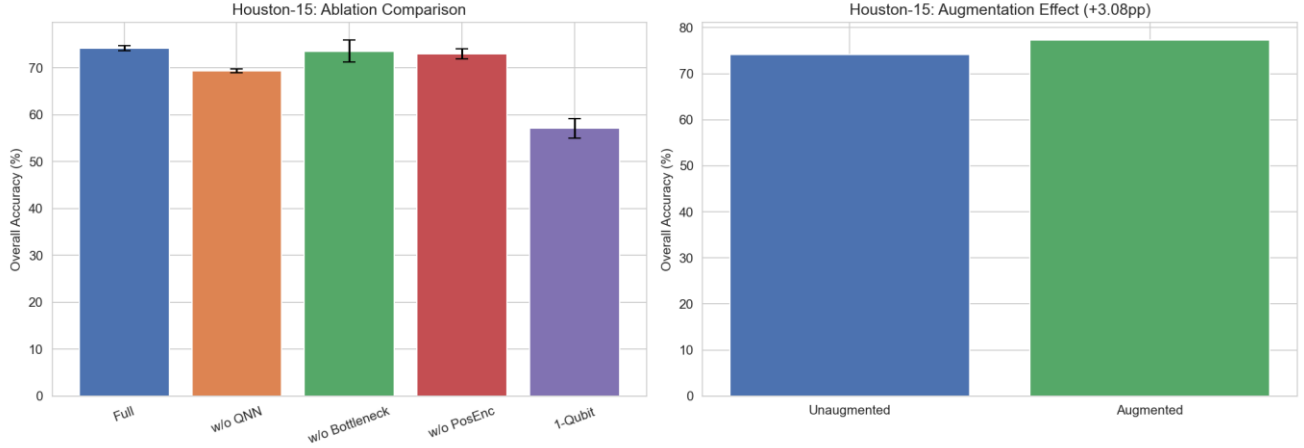
Table 5 (Section 5.2) shows the augmentation effect was smallest, and negative, on Salinas (-0.36 percentage points), the extension dataset with both the largest training set (5,412 patches) and the highest unaugmented accuracy (99.26%), and largest on Houston 2013 (+3.08 points), the dataset with the lowest unaugmented accuracy (74.19%).

The dataset with the largest augmentation benefit, Houston 2013, does not have the smallest training set among the four extension datasets; KSC (521 patches) and Botswana (324 patches) both have fewer training samples than Houston's 2,535. Ranking the four datasets by unaugmented accuracy instead of by training-set size produces a monotonic relationship with the augmentation effect: Salinas (99.26% OA, -0.36 points), KSC (96.47%, +1.09), Botswana (94.49%, +1.59), and Houston 2013 (74.19%, +3.08). This suggests augmentation's benefit tracks how far a dataset is from ceiling accuracy at least as closely as it tracks training-set size, though the two are related in this project, since the smaller-training-set datasets also tend to be the harder classification problems.

A probe testing the interaction between augmentation and positional encoding on Salinas was planned but not run in this project; this is noted as a limitation in Section 9.



Effect of augmentation on overall accuracy, Salinas, KSC, and Botswana.



Houston 2013 (15-class) ablation comparison and augmentation effect.

7. Efficiency Analysis

Table 11 gives the trainable parameter count for the Full and w/o QNN configurations on each dataset, and Table 12 gives per-sample inference time for the Full configuration.

Dataset	Full (params)	w/o QNN (params)	Difference
Indian Pines	34,984	34,960	24
Pavia University	34,445	34,421	24
Salinas	34,900	34,876	24
KSC	34,705	34,681	24
Botswana	34,770	34,746	24
Houston 2013 (15-class)	34,835	34,811	24

Table 11. Trainable parameter count, Full versus w/o QNN configuration.

Dataset	Inference time (ms/sample)
Indian Pines	0.2983
Pavia University	0.2787
Salinas	0.2775
KSC	0.2803
Botswana	0.2783
Houston 2013 (15-class)	0.2765

Table 12. Per-sample inference time, Full configuration, architecture-level timing on untrained weights.

The quantum circuit contributes 24 trainable parameters out of roughly 35,000 total on every dataset, since only the 4-qubit circuit's own weights (2 StronglyEntanglingLayers, 4 qubits, 3 rotation parameters per qubit per layer) differ between the Full and w/o QNN configurations; the rest of the architecture is identical. This is consistent with the original paper's own framing of parameter efficiency as its main contribution rather than an accuracy advantage.

Inference time was measured on freshly initialized, untrained weights, since forward-pass compute cost depends on tensor shapes and architecture rather than on learned weight values; these figures should be read as architecture-level timing, not as a property of any specific trained model. Per-sample inference time was effectively constant across datasets (0.2765 to 0.2983 milliseconds per sample), consistent with

the architecture's shape being nearly identical across datasets, since only the PCA input dimension k differs and the rest of the network (a single transformer layer processing 225 tokens of width 64) does not depend on the dataset.

8. Discussion

Across the five datasets that received the full ablation suite, the width of the token-encoder bottleneck was consistently the dominant factor in accuracy, not the choice between a quantum circuit and a classical projection of the same width. Removing the bottleneck entirely (w/o Bottleneck) improved accuracy on every dataset, by margins as large as 7.46 percentage points on Indian Pines; reducing the quantum circuit to a single qubit degraded accuracy severely on every dataset, by margins as large as 20.82 points. The direct quantum-versus-classical comparison (w/o QNN) was small on four of five datasets and reached practical significance only on Indian Pines, where the classical mirror won.

The full 15-class Houston 2013 scene showed the largest quantum-favoring margin observed in this project: the quantum encoder outperformed its classical mirror by 4.83 percentage points on the hardest and lowest-accuracy dataset tested. This is a larger margin, in either direction, than the quantum-versus-classical comparison produced on any other dataset, including Indian Pines, where the same comparison favored the classical mirror by 3.44 points. A per-class check found this advantage concentrated in a subset of classes rather than spread evenly, and not fully stable across seeds for two of the four classes examined. This project tested the difficulty-dependence hypothesis on only one hard dataset and cannot generalize it beyond that observation.

The embedding-separability results (Section 6.4) do not uniformly support the original paper's claim that its quantum encoder produces a more class-discriminative feature space. The quantum embedding space scored higher on two of three separability metrics on Pavia University but not on Indian Pines, and scored lower on linear-probe accuracy on both datasets. A single, unqualified claim of superior discriminability is not supported by this project's measurements on either dataset tested.

Data augmentation's benefit tracked how far a dataset's unaugmented accuracy sat from ceiling more closely than it tracked training-set size alone (Section 6.5), helping most on the hardest dataset (Houston 2013) and slightly hurting on the easiest of the four extension datasets (Salinas).

Taken together, these results are consistent with the original paper's central claim, that its quantum encoder is not necessary for the accuracy this architecture achieves, on most of the datasets and conditions tested in this project, while identifying one dataset and one evaluation angle (embedding separability) where the picture is more mixed than a single reproduction on the paper's original three benchmarks would suggest.

9. Limitations

The following limitations are specific to this project and are not fully resolved within it.

- Indian Pines' 10.11-percentage-point gap against the paper's reported accuracy is diagnosed (per-class sample scarcity, ruled-out training-split-size explanation) but not closed. The original paper provides no public code release and does not report per-seed variance, so the remaining gap

cannot be attributed with confidence to this implementation or to the paper's own evaluation methodology.

- The original paper's 7-class Houston 2013 subset uses a scene definition (954 by 210 pixels, 48 bands) that does not match the official Houston 2013 scene available to this project; it corresponds to a Houston 2013 and Houston 2018 overlap region that would require a dataset not available to this project. This subset was not evaluated.
- A spatially disjoint train-test split probe for Botswana, an activation-function sensitivity check for the w/o QNN classical mirror, and an augmentation-by-positional-encoding interaction probe for Salinas were all planned in this project's protocol but were not run.
- Noise augmentation was applied to PCA-transformed token values rather than to raw normalized spectral bands as originally specified, because the raw normalized patches were not retained after preprocessing. The direction of the augmentation results reported in Sections 5.2 and 6.5 is unlikely to depend on this, since geometric augmentation, which was applied as specified, is the larger component of the augmentation procedure, but the exact magnitude of the noise component's effect was not verified against the original specification.
- Every comparison in the ablation study's McNemar testing (Table 8) reached statistical significance after correction, including comparisons with effect sizes below 0.5 percentage points; this reflects the large size of several test sets (up to 43,304 pixels) rather than uniform practical importance, and the effect-size and magnitude columns should be read alongside the significance columns rather than in isolation.
- A per-class check of Houston 2013's quantum-versus-classical comparison across three seeds found two of four classes examined did not show a consistent direction across seeds (Section 6.3); conclusions in this document about Houston 2013's per-class pattern are limited to the two classes that did show a consistent direction.
- Because model checkpoints were not retained between processing stages, the Full configuration at seed 42 was retrained independently for the headline results, the ablation study, and, for Indian Pines and Pavia University, the artifact-extraction step producing the receiver operating characteristic, precision-recall, calibration, and t-SNE figures. These retrains differ from one another by less than one percentage point in overall accuracy, attributable to GPU floating-point non-determinism, but figures and tables in this document may draw on different specific runs of the same nominal configuration and seed, as detailed in Section 4.
- Training and validation accuracy and loss curves were recorded only for Indian Pines and Pavia University, not for the four extension datasets, due to the additional training runs this would have required.
- The original paper's text available to this project reports only overall accuracy for Pavia University (99.94%); average accuracy and Cohen's kappa are not available for that dataset's fidelity comparison.

10. Conclusion

This project reproduced QuantFormer's architecture and training protocol on Indian Pines and Pavia University, matching the paper's reported accuracy on Pavia University within 1 percentage point while leaving a 10-point gap on Indian Pines only partially explained. Extending the identical pipeline and a broadened ablation design to four datasets the original paper did not evaluate found that the width of the token-encoder bottleneck, not the choice between a quantum circuit and a classical projection, was the dominant factor in accuracy on most datasets tested, consistent with the original paper's own finding on its three benchmarks. The direct quantum-versus-classical comparison itself was negligible on four

datasets and large in magnitude on two: the classical mirror won by 3.44 percentage points on Indian Pines, and the quantum encoder won by 4.83 points, its largest margin in either direction, on the full 15-class Houston 2013 scene, the hardest classification problem in this project. Houston 2013 was also the only dataset where the full quantum configuration outperformed every one of the four ablation configurations rather than losing to the wider, bottleneck-free variant. Embedding-separability measurements gave a mixed answer to the paper's claim of a more discriminative quantum feature space, and a secondary experiment found that data augmentation's benefit tracked problem difficulty. These results support the original paper's central claim under most conditions tested while identifying specific circumstances, a harder classification problem and certain separability metrics, where the comparison between quantum and classical token encoding is closer or reversed.

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Appendix

Table 13 gives the full 16-class per-class breakdown recorded for Indian Pines at seed 42, the only dataset for which this project recorded a complete per-class table in its session log. Per-class tables for the remaining five datasets were computed and saved during this project (macro and weighted precision, recall, and F1, per-class Producer's and User's Accuracy) but are not reproduced here; they are stored in the project's phase7_full_metrics.json output file.

Class	Training samples	Test accuracy
0	5	83.78%
1	143	84.94%
2	83	80.57%
3	24	86.77%
4	48	79.59%
5	73	92.47%
6	3	59.09%
7	48	98.17%
8	2	37.50%
9	97	88.43%
10	245	91.50%

11	59	75.16%
12	20	84.15%
13	126	96.74%
14	39	84.42%
15	9	76.00%

Table 13. Indian Pines per-class training sample count and test accuracy, Full configuration, seed 42.